

ROC curve for the sign-agreement gate. The x-axis is 'false positive rate' and the y-axis is 'true positive rate', both ranging from 0.0 to 1.0. A solid blue line represents the 'sign-agreement gate (AUC 0.91)', which rises sharply from (0,0) to approximately (0.15, 0.95) and then remains near the top. A dashed grey line represents the 'chance' level, running diagonally from (0,0) to (1,1).

The scatter plot displays the ensemble sign-agreement (10 seeds) on the y-axis, ranging from 0.5 to 1.0. The x-axis is divided into two categories: 'sign-correct' and 'sign-wrong'. A dashed horizontal line at 0.9 represents the 'trust threshold 0.9'. A text box indicates that at $\tau=0.9$, 10.6/11 components are certified trustworthy, and the hybrid follows the free gradient, preserving the TMM descent direction (cosine 0.996) at <1 solve/grad. The 'sign-correct' category shows a high density of points at 1.0, with a few points at 0.9, 0.8, 0.6, and 0.5. The 'sign-wrong' category shows a distribution of points, with a cluster around 0.9 and others at 1.0, 0.8, 0.7, 0.6, and 0.5.